New York University Response to the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO) RFI on the Development of an Artificial Intelligence (AI) Action Plan

NYU is a world-renowned research institution that is committed to advancing knowledge and solving some of the world's most pressing challenges. Our rapidly expanding and diverse research portfolio has significantly contributed to various fields, including medicine, engineering, mathematics, computing, data science, social sciences, humanities, and education.

Since its founding in 1831, NYU has been a global leader in higher education and is now the largest private university in the United States. We are also one of the world's most prestigious and respected research universities, with an annual research budget of $1.5 billion and a reputation for being a top producer of patents and licensing revenue among US universities. We are home to nearly 30,000 undergraduate students, over 27,000 graduate and professional students, and 2,247 tenure-track and tenured faculty members.

At NYU, we are passionate about leveraging our research initiatives to ensure results and societal impact. We firmly believe that innovation is essential to American competitiveness, prosperity, health, and security and that universities play a critical and fundamental role in driving that innovation. We have established a robust entrepreneurial ecosystem with all the resources and support systems our faculty, researchers, and students need to propel their research out of the lab and into the real world. For example, NYU fosters strong research collaborations with top technology, media, and financial firms, including Google, Facebook/MEta, IBM, Microsoft, AT&T, Bell Labs, NEC, Disney, Lucas Films, Samsung, Qualcomm, Amazon, and many more. With our location in the heart of the startup scene in NYC, NYU is also ranked 8th among universities nationally for generating new startups, and we were ranked #10 in the U.S. for entrepreneurship & innovation by Fast Company.

As AI has advanced, its potential now reaches across broad swathes of the economy and many areas of research. To realize the transformative potential of this technology will require support for foundational research, standards, and testing to ensure reliability and usability, actions targeted at essential and promising applications, infrastructure, and education and workforce. Please see the following pages for more details on each of these key areas.

Research and Development Needs

Continued robust federal funding for academic research and educational programs is critical to ensuring that the U.S. stays at the forefront of AI advancement. Investments in fundamental research to develop AI technologies through core programs such as those at the National Science Foundation, Department of Defense, and Department of Energy should continue. For many years, NYU has been active in areas of AI and machine learning for experimentation and research. For example, NYU's Courant Institute of Mathematical Sciences serves as a leading center for research and advanced training in computer science and mathematics. One of the hallmarks of the Institute, which should be reflected in the AI plan, is the support of an interdisciplinary approach, fostering collaboration between mathematics, computer science, and engineering for AI research and techniques that look ahead of where industry is focused today and focus on next-generation systems and applications. Industry looks to and partners with universities like NYU to generate and de-risk innovative ideas that will lead to future commercial breakthroughs.

NYU faculty research expertise and strengths cover several key areas underpinning AI including algorithms and theory of computation, formal methods and verification, machine learning, natural language and speech processing, and scientific computing. As just one example, natural language processing has had a long history at NYU. The Linguistic String Project was one of the pioneers in natural language processing research in the U.S.

Areas requiring research funding include:

- Data harmonization and normalization techniques, such as data cleaning, validation, and error correction
- Analysis of highly distributed data sources, including methods for tracking data provenance, coping with sampling biases and heterogeneity, ensuring data integrity, privacy, security, and sharing, and visualizing massive datasets
- Intersection of modeling and AI
- Explainable AI models
- Creative AI for when you need creativity and reliable AI for when you need accuracy, a specific forecast, scientific result, or otherwise concrete answer

Beyond these foundational efforts, research agencies should support teams to pursue key application areas to connect AI to science, health, energy, defense, space, and other agency missions. For example, the NSF AI Research Institutes should continue to be expanded to new topics and competitions to cover promising use-inspired research and focused AI applications.

AI for Health

Health is a particular area of promise for AI research and the National Institutes of Health (NIH), Advanced Research Projects Agency for Health (ARPA-H), and other parts of the Department of Health and Human Services (HHS) should look for opportunities to expand their work in AI. Health research has lagged other areas in AI advances and adoption, and the opportunity to develop better diagnostics, personalized treatments, new efficiencies for hospitals and the health care system, and address chronic disease is enormous. AI may also lead to better methods for reproducibility and testing of biomedical results to enable faster and more efficient and effective scale up for new treatments and cures. NIH should also look to partner with NSF to enable health and biomedical-focused AI Research Institutes.

AI Evaluation and Reliability

To truly unleash AI innovation and adoption, users must be able to trust and rely on AI and ensure it is as free as possible from security vulnerabilities. Testing and voluntary standards are helpful to achieve these goals and can be done so in ways that do not overburden industry. NIST has decades of experience creating industry consortiums where data can be shared and standards can be developed that help all participating members with pre-competitive activities. The NIST

AI Safety Institute has the potential to serve this role. NYU is part of the nascent NIST AI Safety Research Consortium, which was started in recent years but has not been used to its full potential. The Administration should use the AI Safety Institute to address reliability and security, test for vulnerabilities, and develop needed standards. The Research Consortium can be reimagined as an avenue to bring together industry and academic partners to share data and information and enable research that informs the Institute's goals. Putting a real agenda forward for the Consortium with some incentives for participation would significantly enhance its effectiveness and create an innovation-friendly mechanism for testing AI that unleashes its potential.

At NYU, we take research security very seriously. As you consider AI policies, the current research security approach of the federal government reaches the right balance, enabling the government to address security risks to research while ensuring a continued innovation ecosystem where the exchange of ideas and research is essential to our competitiveness and leadership.

Infrastructure

Federal infrastructure is needed to ensure AI research can be done by academics, small businesses, and others that do not have enormous AI resources. In New York, we have developed Empire AI, which can be a model for national shared investment. The NSF *National AI Research Resource* (NAIRR) pilot is underway, and NYU was glad to see this effort called out specifically in the Trump Administration's AI Executive Order. NSF should launch the full NAIRR to revolutionize AI capabilities and enable a flourishing research and innovation ecosystem.

An open ecosystem for AI research is crucial for fostering collaboration, accelerating innovation, and ensuring that the benefits of AI are maximally distributed. The United States must continue to recognize comparative scientific strengths worldwide and utilize our alliances and global partnerships to bolster our scientific ecosystems and enhance our AI capabilities.

Education and Workforce

In addition to investments in fundamental research in AI, prioritizing a coordinated strategy to invest in AI research, development, integration, and implementation in education and training is crucial to advancing our leadership in STEM and AI educational excellence. No national K-12 AI curriculum or policy exists, and most American teachers have received little to no training on AI tools, leaving them understandably hesitant. Surveys show that only 30% of U.S. teachers feel moderately confident using AI in their classrooms, while nearly half admit to being largely unprepared. Federal support for AI in education has been limited to isolated grants and guidelines, without the cohesive, well-funded strategy seen abroad. Challenges will arise while creating and integrating a more effective AI policy. It is imperative that collaboration between federal and state leaders, teachers and tech developers, researchers and practitioners, and other relevant organizations across communities is at the foundation of creating common and agreed-upon AI policies. The Institute for Education Sciences and the NSF should continue to work together to address AI challenges and applications for education. Critically, research and other investments are also needed to ensure that students learn and develop the skills they need to work and be competitive in an AI-infused future, including teaching creativity so that students can collaborate effectively with AI-powered tools. Example investments could include support to develop AI-powered teacher assistant tools to help address workforce shortages, investments to bolster university and K-12 school district partnerships to test AI interventions, and support for teacher professional development programs that provide training on AI and teaching best practices.

Alongside work to build AI literacy and skills, efforts should ensure we have the talent pathways and pipeline to fuel our continued leadership in AI research and development. Programs such as the NSF Research Traineeships, Graduate Research Fellowships, and Research Experiences for Undergraduates; Department of Education Graduate Assistance in Areas of National Need (GAANN) program; and Department of Energy Computational Science Graduate Fellowship are critical to encourage domestic STEM students to pursue graduate education and gain exposure to new challenges and skills. These investments are needed to train the next generation and ensure our future competitiveness. Investment in AI education across all levels, from K-12 to graduate and post-graduate, is necessary to increase the domestic workforce and talent pipeline developing new AI approaches and tools and implementing them across sectors and application areas.

Upskilling investments are also helpful in transitioning current workers and ensuring they are not left behind as AI tools expand. NYU Future Labs are a potential model for upskilling, and we are committed to working with community college partners, such as the City College of New York, to enhance workforce training.